

HEYANG YUAN

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EDUCATION

York University, Toronto, ON, Canada Faculty of Liberal Arts & Professional Studies Master of Information Systems and Technology Supervised by Zhenhao Li	Sept 2025 - Present
York University, Toronto, ON, Canada Lassonde School of Engineering Department Honour Bachelor of Computer Science	Sept 2019 - April 2024

PROFESSIONAL EXPERIENCE

York University, Toronto, ON, Canada Research Assistant Supervised by Meiyang Qin (York University). Working on AI/AR interaction frameworks	April 2025 - Sept 2025
Fortrip AI, Toronto, ON, Canada Software Developer Internship	June 2025 - Sept 2025
Easy Trip. Inc, Toronto, ON, Canada Software Developer Internship	Feb 2025 - June 2025
York University. Inc, Toronto, ON, Canada Undergraduate Researcher (Full-Time Contract) Supervised by Aijun An (York University + NHK). Working on NLP models for sign language processing	Sept 2023 - Dec 2023

RESEARCH EXPERIENCE

Coding Agents for Non-Functional Improvements Benchmarking - Integrated Test-Driven Development principles into the evaluation pipeline, utilizing test suites to ensure the functional integrity of agent-generated code. - Developed a benchmark suite with executable evaluation rules to assess coding agents on non-functional improvements. - Designed an evaluation framework to measure agentic performance in complex, long-horizon software maintenance tasks.	May 2026 - July 2026
Reinforcement Learning in the Software Development Lifecycle - Conducted a survey of primary studies, mapping RL applications across the seven SDLC phases. - Proposed a taxonomy of the seven SDLC phases has been applied, showing the shift from testing-centric work toward broader lifecycle coverage. - A systematic examination of reward modeling strategies across the SDLC, summarizing common patterns, parameters, and trade-offs to guide future design choices and identify promising research directions.	March 2026 - May 2026
Large Language Models for Logging Code Security - Derive a taxonomy of logging code security issues synthesized from literature and vulnerability databases. - Constructed SecLogging, a benchmark dataset for systematic evaluation of LLM-based detection and repair of logging vulnerabilities.	Sept 2025 - Feb 2026

- Conduct a systematic evaluation on LLMs, provide actionable insights for practitioners and researchers on secure logging practices.

RESEARCH COMPETENCIES

- **Agentic Systems & RL Design:** Research experience in analyzing autonomous agent frameworks for software engineering, examining Markov Decision Processes and reinforcement learning to support goal-oriented decision-making.
- **Reward Engineering & Alignment:** Studied reward modeling for LLM-based agents, analyzing PPO, DPO, and RLHF to enhance semantic consistency and align agent behaviors with complex development objectives.
- **LLM for Software Engineering:** Focused on applying LLMs to code-centric tasks, including automated program repair, vulnerability detection, and security-aware analysis.
- **Agentic Evaluation Frameworks:** Developed end-to-end evaluation pipelines, utilizing execution-based feedback and formal verification to measure the robustness and non-functional performance of intelligent agents.

RESEARCH INTERESTS

Intelligent Software engineering (AI4SE), specifically,

- **Intelligent Code Analysis:** Automated Vulnerability Detection, Security-aware Code Analysis
- **Software Reliability:** Automatic Program Repair, Reward Engineering for Coding Agents

PUBLICATIONS

1. “NFICodeBench: Benchmarking Coding Agents for Non-Functional Improvements Beyond Functional Correctness” [ICSE 2027]
Pengyu Xue, **He Yang Yuan***, Xin Wang, Junkai Chen, Haonan Zhang, Boyuan Chen, Zishuo Ding, Zhenhao Li, Weiye Shang
Submitted to *49th IEEE/ACM International Conference on Software Engineering (ICSE 2027, Research Track)* (Under Review)
2. “Reinforcement Learning in the Software Development Lifecycle: Methods, Applications, and Reward Design” [TSE 2026]
He Yang Yuan*, Xin Wang, Pengyu Xue, Zhenhao Li, Zishuo Ding, Boyuan Chen, Dayi Lin, Weiye Shang
Submitted to *IEEE Transactions on Software Engineering (TSE)*, 2026. (Under Review)
3. “Towards Secure Logging: Characterizing and Benchmarking Logging Code Security Issues with LLMs” [FSE 2026]
He Yang Yuan*, Xin Wang, Kundi Yao, An Ran Chen, Zishuo Ding, Zhenhao Li
ACM International Conference on the Foundations of Software Engineering (ESEC/FSE 2026), Montreal, QC, Canada, July 2026.

TEACHING EXPERIENCE

ITEC 3020 Web Technology

[Sept 2025 - Present]

Teaching Assistant

- Mentored students on academic success strategies, including effective time management, study planning, and project prioritization to achieve high performance.
- Designed and evaluated assessment rubrics for homework assignments, ensuring consistent and fair grading standards.